GCP BIGQUERY - ASSIGNMENT

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Batch: 2027

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Creating Database – Eccomerce Database with Table Entries

Query:

```
CREATE SCHEMA IF NOT EXISTS ecommerce_dataset

OPTIONS(location = 'US');


CREATE OR REPLACE TABLE ecommerce_dataset.orders AS


SELECT 'ORD1001' AS Order_id, 'CUST101' AS Customer_id, 'P101' AS
Product_id, DATE '2026-08-05' AS Order_Date, 'Laptop' AS Product, 2 AS
Quantity, 55000 AS Unit_Price, 'South' AS Region, 'Delivered' AS Status

UNION ALL

SELECT 'ORD1002', 'CUST102', 'P102', DATE '2026-08-12', 'Smartphone', 1,
25000, 'North', 'Delivered'

UNION ALL

SELECT 'ORD1003', 'CUST103', 'P103', DATE '2026-08-20', 'Headphones', 3,
2000, 'East', 'Completed'

UNION ALL

SELECT 'ORD1004', 'CUST101', 'P104', DATE '2026-09-03', 'Keyboard', 2, 1500,
'South', 'Delivered'

UNION ALL

SELECT 'ORD1005', 'CUST104', 'P101', DATE '2026-09-10', 'Laptop', 1, 55000,
'West', 'Delivered'

UNION ALL

SELECT 'ORD1006', 'CUST105', 'P105', DATE '2026-09-15', 'Monitor', 2, 12000,
'North', 'Completed'

UNION ALL

SELECT 'ORD1007', 'CUST102', 'P103', DATE '2026-09-22', 'Headphones', 2,
2000, 'North', 'Delivered'
```

```
UNION ALL

SELECT 'ORD1008', 'CUST106', 'P102', DATE '2026-10-02', 'Smartphone', 2, 25000, 'East', 'Delivered'

UNION ALL

SELECT 'ORD1009', 'CUST103', 'P106', DATE '2026-10-05', 'Mouse', 4, 800, 'South', 'Completed'

UNION ALL

SELECT 'ORD1010', 'CUST101', 'P101', DATE '2026-10-10', 'Laptop', 1, 55000, 'South', 'Delivered'

UNION ALL

SELECT 'ORD1011', 'CUST104', 'P104', DATE '2026-10-12', 'Keyboard', 3, 1500, 'West', 'Pending'

UNION ALL

SELECT 'ORD1012', 'CUST105', 'P105', DATE '2026-10-15', 'Monitor', 1, 12000, 'North', 'Delivered'

UNION ALL

SELECT 'ORD1013', 'CUST102', 'P102', DATE '2026-10-18', 'Smartphone', 1, 25000, 'North', 'Delivered'

UNION ALL

SELECT 'ORD1014', 'CUST107', 'P107', DATE '2026-10-21', 'Printer', 2, 9000, 'East', 'Cancelled'

UNION ALL

SELECT 'ORD1015', 'CUST103', 'P103', DATE '2026-10-25', 'Headphones', 5, 2000, 'South', 'Delivered';
```

Output:

Order_id	Customer_	Product_ic	Order_Dat	Product	Quantity	Unit_Price	Region	Status
ORD1004	CUST101	P104	03/09/2026	Keyboard	2	1500	South	Delivered
ORD1005	CUST104	P101	10/09/2026	Laptop	1	55000	West	Delivered
ORD1006	CUST105	P105	15/09/2026	Monitor	2	12000	North	Completed
ORD1007	CUST102	P103	22/09/2026	Headphones	2	2000	North	Delivered
ORD1008	CUST106	P102	02/10/2026	Smartphone	2	25000	East	Delivered
ORD1009	CUST103	P106	05/10/2026	Mouse	4	800	South	Completed
ORD1010	CUST101	P101	10/10/2026	Laptop	1	55000	South	Delivered
ORD1011	CUST104	P104	12/10/2026	Keyboard	3	1500	West	Pending
ORD1012	CUST105	P105	15/10/2026	Monitor	1	12000	North	Delivered
ORD1013	CUST102	P102	18/10/2026	Smartphone	1	25000	North	Delivered
ORD1014	CUST107	P107	21/10/2026	Printer	2	9000	East	Cancelled
ORD1015	CUST103	P103	25/10/2026	Headphones	5	2000	South	Delivered

<u>TASK 01</u>: Remove the Duplicate Orders

Query:

```
SELECT *
FROM ecommerce_dataset.orders
QUALIFY ROW_NUMBER() OVER (
  PARTITION BY Order_id
  ORDER BY Order_Date
) = 1;
```

Output:

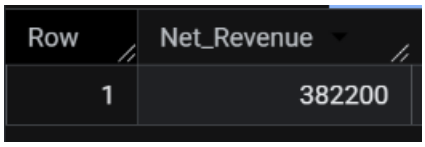
Order_id	Customer_	Product_ic	Order_Dat	Product	Quantity	Unit_Price	Region	Status
ORD1001	CUST101	P101	05/08/2026	Laptop	2	55000	South	Delivered
ORD1002	CUST102	P102	12/08/2026	Smartphone	1	25000	North	Delivered
ORD1003	CUST103	P103	20/08/2026	Headphones	3	2000	East	Completed
ORD1004	CUST101	P104	03/09/2026	Keyboard	2	1500	South	Delivered
ORD1005	CUST104	P101	10/09/2026	Laptop	1	55000	West	Delivered
ORD1006	CUST105	P105	15/09/2026	Monitor	2	12000	North	Completed
ORD1007	CUST102	P103	22/09/2026	Headphones	2	2000	North	Delivered
ORD1008	CUST106	P102	02/10/2026	Smartphone	2	25000	East	Delivered
ORD1009	CUST103	P106	05/10/2026	Mouse	4	800	South	Completed
ORD1010	CUST101	P101	10/10/2026	Laptop	1	55000	South	Delivered
ORD1011	CUST104	P104	12/10/2026	Keyboard	3	1500	West	Pending
ORD1012	CUST105	P105	15/10/2026	Monitor	1	12000	North	Delivered
ORD1013	CUST102	P102	18/10/2026	Smartphone	1	25000	North	Delivered
ORD1014	CUST107	P107	21/10/2026	Printer	2	9000	East	Cancelled
ORD1015	CUST103	P103	25/10/2026	Headphones	5	2000	South	Delivered

TASK 2: Find the Net Revenue

Query:

```
SELECT SUM(Quantity * Unit_Price) AS Net_Revenue
FROM ecommerce_dataset.orders
WHERE Status IN ('Delivered', 'Completed');
```

Output:



Row	Net_Revenue
1	382200

TASK 03: Find the Top 3 Products in each Region

Query:

```
SELECT Region, Product, Revenue, Product_Rank
FROM (
  SELECT Region, Product,
        SUM(Quantity * Unit_Price) AS Revenue,
        RANK() OVER (
          PARTITION BY Region
          ORDER BY SUM(Quantity * Unit_Price) DESC
        ) AS Product_Rank
  FROM ecommerce_dataset.orders
  WHERE Status IN ('Delivered', 'Completed')
  GROUP BY Region, Product
)
```

```
WHERE Product_Rank <= 3
ORDER BY Region, Product_Rank;
```

Output:

Job information | Results | Visualisation | JSON | Execution details | Execution graph

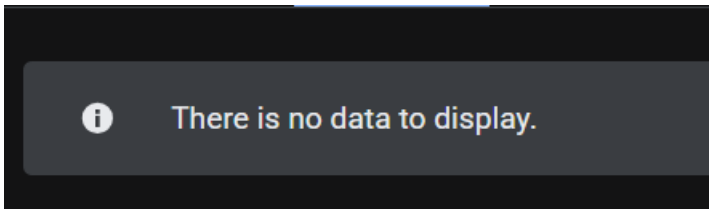
Row	Region	Product	Revenue	Product_Rank
1	East	Smartphone	50000	1
2	East	Headphones	6000	2
3	North	Smartphone	50000	1
4	North	Monitor	36000	2
5	North	Headphones	4000	3
6	South	Laptop	165000	1
7	South	Headphones	10000	2
8	South	Mouse	3200	3
9	West	Laptop	55000	1

TASK 4: Identify Inactive Customers

Query:

```
SELECT Customer_id,
    MAX(Order_Date) AS Last_Purchase_Date
FROM ecommerce_dataset.orders
WHERE Status IN ('Delivered', 'Completed')
GROUP BY Customer_id
HAVING MAX(Order_Date) < DATE_SUB(
  CURRENT_DATE(), INTERVAL 90 DAY
);
```

Output:



TASK 5: Find the Customers who Purchase Continuously 3 Months

Query:

```
SELECT Customer_id,
      COUNT(DISTINCT FORMAT_DATE('%Y-%m', Order_Date))
       AS Months_Purchased
FROM ecommerce_dataset.orders
WHERE Status IN ('Delivered', 'Completed')
  AND Order_Date >= DATE '2026-08-01'
  AND Order_Date < DATE '2026-11-01'
GROUP BY Customer_id
HAVING COUNT(DISTINCT FORMAT_DATE('%Y-%m', Order_Date)) = 3;
```

Output:

Row	Customer_id	Months_Purchased
1	CUST102	3
2	CUST101	3

TASK 6: Build a Customer Lifetime Value Report

Query:

```
SELECT Customer_id,
    SUM(Quantity * Unit_Price) AS Lifetime_Revenue,
    COUNT(DISTINCT Order_id) AS Total_Orders
FROM ecommerce_dataset.orders
WHERE Status IN ('Delivered', 'Completed')
GROUP BY Customer_id
ORDER BY Lifetime_Revenue DESC;
```

Output:

Row	Customer_id	Lifetime_Revenue	Total_Orders
1	CUST101	168000	3
2	CUST104	55000	1
3	CUST102	54000	3
4	CUST106	50000	1
5	CUST105	36000	2
6	CUST103	19200	3

TASK 7: Find the Monthly Sales

Query:

```
SELECT FORMAT_DATE('%Y-%m', Order_Date) AS Sales_Month,
    SUM(Quantity * Unit_Price) AS Monthly_Sales
FROM ecommerce_dataset.orders
WHERE Status IN ('Delivered', 'Completed')
GROUP BY Sales_Month
```

```
ORDER BY Sales_Month;
```

Output:

Row	Sales_Month	Monthly_Sales
1	2026-08	141000
2	2026-09	86000
3	2026-10	155200

TASK 8: Find the Unit Price Below 10000 in a Particular Month

Query:

```
SELECT Product, Unit_Price, Order_Date, Status
FROM ecommerce_dataset.orders
WHERE Unit_Price < 10000
  AND Order_Date >= DATE '2026-10-01'
  AND Order_Date < DATE '2026-11-01'
ORDER BY Unit_Price;
```

Output:

Row	Product	Unit_Price	Order_Date	Status
1	Mouse	800	2026-10-05	Completed
2	Keyboard	1500	2026-10-12	Pending
3	Headphones	2000	2026-10-25	Delivered
4	Printer	9000	2026-10-21	Cancelled

TASK 09: Find the Customer who is Purchased Particular Product Multiple Times in 6 Months

Query:

```
SELECT Customer_id,
    COUNT(DISTINCT Order_id) AS Purchase_Count
FROM ecommerce_dataset.orders
WHERE LOWER(Product) = 'laptop'
  AND Status IN ('Delivered', 'Completed')
  AND Order_Date >= DATE_SUB(CURRENT_DATE(), INTERVAL 6 MONTH)
GROUP BY Customer_id
HAVING COUNT(DISTINCT Order_id) >= 2;
```

Output:

Row	Customer_id	Purchase_Count
1	CUST101	2

TASK 10: Find the Total Sales Report for each Product in a Particular Month

Query:

```
SELECT Product,
    SUM(Quantity) AS Total_Quantity,
    SUM(Quantity * Unit_Price) AS Total_Sales
FROM ecommerce_dataset.orders
WHERE Status IN ('Delivered', 'Completed')
  AND Order_Date >= DATE '2026-10-01'
  AND Order_Date < DATE '2026-11-01'
```

GROUP BY Product

ORDER BY Total_Sales DESC;

Output:

Row	Product	Total_Quantity	Total_Sales
1	Smartphone	3	75000
2	Laptop	1	55000
3	Monitor	1	12000
4	Headphones	5	10000
5	Mouse	4	3200